

Lectures On Class Field Theory

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1 Un peu d'histoire

We start with some motivation: this course will be about number fields (i.e. finite extensions $K|\mathbf{Q}$) and their absolute Galois groups $G_K = \text{Gal}(\overline{K}|K)$, where $\overline{K}$ is a fixed algebraic closure of K . Very little is known about the structure of such automorphism groups in general, and a large chunk of modern number theory/arithmetic geometry is devoted to studying them, e.g. via their representations. In some sense, class field theory consists in the study of its 1-dimensional representations; more precisely, it describes the abelianization $G_K^{\text{ab}} = G_K/[G_K, G_K]$ of the absolute Galois group (equivalently, of its abelian quotients).

1.1 Reciprocity laws

Let p be a prime number, and let a be an integer coprime to p . The *Legendre symbol* $\left(\frac{a}{p}\right)$ is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square modulo } p, \\ -1 & \text{if } a \text{ is not a square modulo } p. \end{cases}$$

Euler proved that $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \pmod{p}$, and **Gauss** proved the following law:

Theorem 1.1 (Quadratic reciprocity). Let $a \neq b$ be different odd primes. Then,

$$\left(\frac{a}{b}\right) \cdot \left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2} \frac{b-1}{2}}.$$

Moreover, the following subsidiary laws hold:

1. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}};$
2. $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$

These laws, together with its multiplicativity in the parameter a , make it possible to compute the Legendre symbol.

Another way of thinking about class field theory is as an attempt to define a “very general” reciprocity law, from which the quadratic reciprocity law and its higher degree analogues can be deduced. One such generalization is **Hilbert’s** reciprocity law: recall that $\mathbf{Q}$ embeds into the p -adic field $\mathbf{Q}_p$ for every prime p , and also into the real numbers $\mathbf{Q}_\infty := \mathbf{R}$. For every prime p (including $p = \infty$) and for all $a, b \in \mathbf{Q}^\times$, we define the *Hilbert symbol* $(a, b)_p$ as

$$(a, b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ has a solution in } \mathbf{Q}_p^2, \\ -1 & \text{otherwise.} \end{cases}$$

Recall that the norm $N : \mathbf{Q}_p(\sqrt{a}) \rightarrow \mathbf{Q}_p$ is defined as $N(x + y\sqrt{a}) = x^2 - ay^2$ if $a \notin \mathbf{Q}_p^{\times 2}$, and as $N = \text{id}$ otherwise. More generally, if $L|K$ is a finite Galois extension with group G , the norm $N_{L|K} : L \rightarrow K$ is

$$N(x) = \prod_{\sigma \in G} \sigma(x)$$

Lemma 1.2. The following statements are equivalent:

1. $(a, b)_p = 1$;
2. The quadratic equation $z^2 = ax^2 + by^2$ has a non-trivial solution;
3. $a \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{b})$.
4. $b \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{a})$.

Proof. The equivalence $1 \iff 2$ is by definition; and $3 \iff 4$ is clear. For $2 \iff 3$, if $b \in \mathbf{Q}_p^{\times 2}$, then we are done. Otherwise, if $b \notin \mathbf{Q}_p^{\times 2}$, then $N(\alpha + \beta\sqrt{b}) = \alpha^2 - b\beta^2$, and if $a := \alpha^2 - b\beta^2$ we see that $z^2 = ax^2 + by^2$ has a non-trivial solution, and vice versa. $\square$

For $p \neq 2, \infty$ and $a, b \in \mathbf{Z}$, write $a = p^{v_p(a)}u$ and $b = p^{v_p(b)}v$ with $p \nmid uv$. The relation between the Hilbert symbol and the Legendre symbol is

$$(a, b)_p = (-1)^{\frac{p-1}{2}v_p(a)v_p(b)} \left(\frac{u}{p}\right)^{v_p(b)} \left(\frac{v}{p}\right)^{v_p(a)} \quad (1.1)$$

$$= \overline{(-1)^{v_p(a)v_p(b)} a^{v_p(b)} b^{-v_p(a)}}^{(p-1)/2}. \quad (1.2)$$

where the bar denotes reduction modulo p . In particular, if p does not divide a , then $(a, b)_p = \left(\frac{a}{p}\right)^{v_p(b)}$ coincides with the Legendre symbol.

Theorem 1.3 (Hilbert reciprocity law). For all $a, b \in \mathbf{Q}^\times$, we have

$$\prod_p (a, b)_p = 1$$

As for computation, one shows that if p does not divide $2ab$ (including $p = \infty$ for $a, b > 0$), then $(a, b)_p = 1$. Moreover, $(a, b)_2 = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}$ for odd a, b . From this and multiplicativity in the first variable, we immediately deduce the quadratic reciprocity law by taking a, b to be odd primes.

An advantage of this formulation is that it generalizes to number fields K , by considering the completions of K at all of its (finite and infinite) “places”. Hilbert then went on to pose his 9th problem, in which he asked for a generalization of this law to n -th powers in arbitrary number fields; this is what class field theory achieves. We first consider the case of extensions which contain a primitive n -th root of unity:

1.2 Kummer case

Assume now $K|\mathbf{Q}$ is a finite extension such that $\mu_n \subset K$, and let $\mathcal{O}_K$ be its ring of integers. If $P \subset \mathcal{O}_K$ is a prime ideal, we would like to define a symbol $(a, b)_P \in \mu_n$ such that a product formula holds, i.e.

$$\prod_P (a, b)_P = 1$$

if $a, b \in K^\times$. Moreover, for P finite (which is always the case for $n \neq 2$) which does not divide (n) we shall have

$$(a, b)_P = \overline{(-1)^{v_P(a)v_P(b)} a^{v_P(b)} b^{-v_P(a)} (q-1)^{1/n}},$$

where $q = \#(\mathcal{O}_K/P)$. In particular, if P does not divide (a) , then $(a, b)_P = a^{\frac{q-1}{n} v_P(b)} \bmod P$.

So, if $v_P(b) = 1$, then

$$\left(\frac{a}{P}\right) := (a, b)_P$$

does not depend on the choice of b and it is the unique n -th root of unity, that is $a^{\frac{q-1}{n}} \bmod P$; in other words, $\left(\frac{a}{P}\right) = 1$ if and only if $x^n = a \bmod P$ has a solution. Some more properties of this symbol are:

1. $(a, b)_P = 1 \iff b$ is a norm from $K_P(\sqrt[n]{a}) \iff a$ is a norm from $K_P(\sqrt[n]{b})$;
2. $(a, b)_P$ is multiplicative in both variables and anticommutative;

When μ_n is *not* contained in K_P , an “Hilbert” symbol still exists but is longer associated to $a, b \in K^\times$. Suppose for a moment that $\mu_n \subset K_P$. Then, the extension $K_P(\sqrt[n]{a})|K_P$ is Galois for all $a \notin K_P^{\times n}$, and its Galois group is

$$G_a := \text{Gal}(K_P(\sqrt[n]{a})|K_P) \cong \mathbf{Z}/n\mathbf{Z}$$

Then, the group $\text{Hom}(G_a, \mu_n) \cong \mathbf{Z}/n\mathbf{Z}$ is generated by the character

$$\overline{\chi}_a : \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n \cong \mathbf{Z}/n\mathbf{Z}.$$

In general, we shall construct a generalized Hilbert symbol

$$\begin{aligned} \text{Hom}(\text{Gal}(\overline{K_P}|K_P), \mathbf{Z}/n\mathbf{Z}) \times K_P^\times &\rightarrow \mathbf{Z}/n\mathbf{Z}, \\ [\chi, b] &\mapsto (\chi, b)_P \end{aligned}$$

and we'll thus be able to prove the **Artin** reciprocity law: fix some $b \in K^\times$, and suppose that χ is a character in $\text{Hom}(G_K, \mathbf{Z}/n\mathbf{Z})$. Then

$$\sum_P (\chi_P, b) = 0,$$

where χ_P is the restriction of χ to $G_{K_P}^{\text{ab}}$ (recall that it embeds into G_K^{ab}).

The generalized Hilbert symbol defines a reciprocity map

$$K_P^\times \rightarrow G_{K_P}^{\text{ab}},$$

which is “almost” an isomorphism. In particular, for every n we'll have an isomorphism of finite groups

$$K_P^\times/n \rightarrow G_{K_P}^{\text{ab}}/n.$$

This is one of the remarkable features of class field theory, giving a description of the abelian extensions of a field in terms of its arithmetic. To define such a map, it's enough to define the dual map

$$\begin{aligned} \text{Hom}(G_{K_P}^{\text{ab}}/n, \mathbf{Z}/n\mathbf{Z}) &\rightarrow \text{Hom}(K_P^\times/n, \mathbf{Z}/n\mathbf{Z}), \\ [\chi] &\mapsto [b \mapsto (\chi, b)_P]. \end{aligned}$$

2 Computing the Hilbert symbol

Recall the definition: for $a, b \in \mathbf{Q}_p^\times$,

$$(a, b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ is solvable in } \mathbf{Q}_p, \\ -1 & \text{otherwise.} \end{cases}$$

We have seen that

$(a, b)_p = 1 \iff z^2 - ax^2 - by^2$ has a non-trivial solution in $\mathbf{Q}_p^3 \iff b$ is a norm from $\mathbf{Q}_p(\sqrt{a})$, where $N(x + y\sqrt{a}) = x^2 - ay^2$.

Theorem 2.1 (Hilbert reciprocity law). For all $a, b \in \mathbf{Q}^\times$,

$$\prod_p (a, b)_p = 1,$$

where the product runs over all primes p , including $p = \infty$.

Note that the product is finite: $(a, b)_p = 1$ for all $p \nmid 2ab$ (see Lemma 2.4 below).

Lemma 2.2. Let $a, b, c \in \mathbf{Q}_p^\times$. Then

1. $(1, c)_p = 1$;
2. $(-c, c)_p = 1$;
3. if $(a, c)_p = 1$, then $(ab, c)_p = (a, c)_p (b, c)_p$ (this holds in general, see the Corollary below);
4. $(c, c)_p = (-1, c)_p$.

Proof. 1. Consider $N : \mathbf{Q}_p(\sqrt{c}) \rightarrow \mathbf{Q}_p$. Then $N(1) = 1$.

2. If $c \in \mathbf{Q}_p^{\times 2}$, then $\mathbf{Q}_p(\sqrt{c}) = \mathbf{Q}_p$, the norm is the identity and $N(-c) = -c$. If $c \notin \mathbf{Q}_p^{\times 2}$, then $N(\sqrt{c}) = -c$. In both cases $-c$ is a norm from $\mathbf{Q}_p(\sqrt{c})$.

3. If $a = N(x)$ and $b = N(y)$, then $ab = N(x)N(y) = N(xy)$, since the norm is multiplicative by definition.

Moreover, if $a = N(x)$, then $a^{-1} = N(x^{-1})$; if in addition $ab = N(z)$, then

$$b = a^{-1}(ab) = N(x^{-1})N(z) = N(x^{-1}z).$$

So, if $(a, c)_p = 1$ (i.e. $a = N(x)$ is a norm from $\mathbf{Q}_p(\sqrt{c})$), then $(b, c)_p = 1 \iff (ab, c)_p = 1$.

4. By (2) and (3),

$$(c, c)_p = (-c, c)_p (c, c)_p = (-c^2, c)_p = (-1, c)_p,$$

because in general $(ad^2, b)_p = (a, b)_p$. □

Remark 2.3. Since $(ad^2, b)_p = (a, b)_p$, the Hilbert symbol is a well-defined map

$$(\cdot, \cdot)_p : (\mathbf{Q}_p^\times / \mathbf{Q}_p^{\times 2}) \times (\mathbf{Q}_p^\times / \mathbf{Q}_p^{\times 2}) \rightarrow \mu_2.$$

Lemma 2.4. Let $p > 2$. Then $(u, v)_p = 1$ for all $u, v \in \mathbf{Z}_p^\times$. (Recall that $\mathbf{Z}_p$ is local and $\mathbf{Z}_p^\times = \mathbf{Z}_p \setminus (p)\cdot$)

Proof. Let $\bar{u} := u \bmod p$ and $\bar{v} := v \bmod p$; they are non-zero elements of $\mathbf{F}_p$ (units of the local ring). We first show that $1 - \bar{u}x^2 - \bar{v}y^2$ has a solution in $\mathbf{F}_p$. Since $\mathbf{F}_p^\times$ is cyclic of order $p-1$,

$$\#\{x^2 : x \in \mathbf{F}_p\} = \frac{p+1}{2}.$$

Similarly, if $S_{\bar{u}} := \{\bar{u}x^2 : x \in \mathbf{F}_p\}$ and $S_{\bar{v}} := \{1 - \bar{v}y^2 : y \in \mathbf{F}_p\}$, then $\#S_{\bar{u}} = \#S_{\bar{v}} = \frac{p+1}{2}$. Thus, $S_{\bar{u}} \cap S_{\bar{v}} \neq \emptyset$ by the

We conclude by Hensel's lemma: □

Lemma 2.5 (Hensel). Let $f \in \mathbf{Z}_p[X]$ and $a_0 \in \mathbf{Z}_p$ with $f(a_0) \equiv 0 \pmod{p}$ and $f'(a_0) \not\equiv 0 \pmod{p}$. Then there exists $a \in \mathbf{Z}_p$ with $f(a) = 0$ and $a \equiv a_0 \pmod{p}$.

This allows us to lift a solution mod p to a solution in $\mathbf{Z}_p$ (to apply it in the proof above, we have to fix one variable).

Proposition 2.6. Let $p > 2$ and $a, b \in \mathbf{Q}_p^\times$, with $a = p^\alpha u$, $b = p^\beta v$ and $u, v \in \mathbf{Z}_p^\times$. Then

$$(a, b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\bar{u}}{p}\right)^{-\beta} \left(\frac{\bar{v}}{p}\right)^{\alpha}.$$

Proof. Since $(a, b)_p$ depends only on $a, b \bmod \mathbf{Q}_p^{\times 2}$, we may assume $\alpha, \beta \in \{0, 1\}$.

- $\alpha = \beta = 0$: this is Lemma 2.4.
- $\alpha = 1, \beta = 0$: we need $(pu, v)_p = \left(\frac{\bar{v}}{p}\right)$. By Lemma 2.4 we have $(u^{-1}, v)_p = 1$, hence by Lemma 2.2(3)

$$(pu, v)_p = (pu, v)_p \underbrace{(u^{-1}, v)_p}_{=1} = (p, v)_p.$$

If $v \in \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = (p, 1)_p = 1 = \left(\frac{\bar{v}}{p}\right)$ (note that $\bar{v}$ is then also a square mod p).
 If $v \notin \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = -1$, because $z^2 - px^2 - vy^2$ has no non-trivial solution mod p , hence none in $\mathbf{Q}_p$ by homogeneity; also $\left(\frac{\bar{v}}{p}\right) = -1$ (by Hensel, since $v \notin \mathbf{Q}_p^{\times 2}$).

- $\alpha = \beta = 1$: we need $(pu, pv)_p = (-1)^{\frac{p-1}{2}} \left(\frac{\bar{u}}{p}\right) \left(\frac{\bar{v}}{p}\right)$. By Lemma 2.2(2) we have $(-pv, pv)_p = 1$, so by Lemma 2.2(3)

$$(pu, pv)_p = (pu, pv)_p \underbrace{(-pv, pv)_p}_{=1} = (-p^2 uv, pv)_p = (-uv, pv)_p = (pv, -uv)_p,$$

and now we are in the previous case, which gives

$$(pv, -uv)_p = \left(\frac{-uv}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{\bar{u}}{p}\right) \left(\frac{\bar{v}}{p}\right),$$

where $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$, and we used that the Legendre symbol is multiplicative. $\square$

Corollary 2.7. Let $p > 2$. Then $(a, c)_p(b, c)_p = (ab, c)_p$ for all $a, b, c \in \mathbf{Q}_p^\times$. In particular, $(\cdot, \cdot)_p$ is bi-multiplicative.

Proof. Write $a = p^\alpha u$, $b = p^\beta v$ and $c = p^\gamma w$ with $u, v, w \in \mathbf{Z}_p^\times$. By Proposition 2.6,

$$\begin{aligned} (a, c)_p(b, c)_p &= (-1)^{\frac{p-1}{2}\alpha\gamma} \left(\frac{\bar{u}}{p}\right)^\gamma \left(\frac{\bar{w}}{p}\right)^\alpha \cdot (-1)^{\frac{p-1}{2}\beta\gamma} \left(\frac{\bar{v}}{p}\right)^\gamma \left(\frac{\bar{w}}{p}\right)^\beta \\ &= (-1)^{\frac{p-1}{2}\gamma(\alpha+\beta)} \left(\frac{\bar{uv}}{p}\right)^\gamma \left(\frac{\bar{w}}{p}\right)^{\alpha+\beta} = (ab, c)_p. \end{aligned} \quad \square$$

Case $p = 2$. If $a = 2^\alpha u$ and $b = 2^\beta v$ with $u, v \in \mathbf{Z}_2^\times$, define

$$\varepsilon(u) := \frac{u-1}{2} \bmod 2, \quad \omega(u) := \frac{u^2-1}{8} \bmod 2.$$

Then

$$(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(v) + \alpha\omega(v) + \beta\omega(u)}.$$

(Proof not given.) In particular, $(\cdot, \cdot)_p$ is bi-multiplicative for every p .

Proof of Theorem 2.1. Using bi-multiplicativity, we reduce to the case where each of a, b is either -1 or a prime.

- Case $a = -1$: homework.
- Case $a = b = q$ prime: $(q, q)_p = (-1, q)_p$ by Lemma 2.2(4), so we are reduced to the previous case.
- Case $a = 2$, $b > 2$ prime: $(2, b)_p = 1$ for $p \notin \{2, b\}$ (also for $p = \infty$, as $2, b > 0$). Moreover, by the $p = 2$ formula,

$$(2, b)_2 = (-1)^{\omega(b)},$$

and by Proposition 2.6

$$(2, b)_b = \left(\frac{2}{b}\right) = (-1)^{\frac{b^2-1}{8}}.$$

The product of these two factors is 1.

- Case $a \neq b$ odd primes: $(a, b)_p = 1$ for $p \notin \{2, a, b\}$ (as well as $p = \infty$), and

$$(a, b)_p = \begin{cases} (-1)^{\frac{a-1}{2} \frac{b-1}{2}} & p = 2, \\ \left(\frac{a}{b}\right) & p = b, \\ \left(\frac{b}{a}\right) & p = a. \end{cases}$$

Hence the product is 1 by quadratic reciprocity. $\square$

3 Cohomology of groups

Already for the local part of theory we'll require some input from homological algebra, mainly the description of the first cohomology group $H^1(G, A)$ for G some Galois group, and its interpretation in terms of crossed homomorphisms. A concise reference for this material can be found in the notes [2]. We recall some results here for the reader's convenience:

Theorem 3.1 (Cohomology of cyclic groups). Let G be a finite cyclic group of order n , generated by σ . Then,

$$\begin{cases} H^0(G, A) = A^G, \\ H^1(G, A) = A/(1 - \sigma)A, \\ H^2(G, A) = A^G/NA, \\ H^i(G, A) \cong H^{i+2}(G, A) \text{ for all } i \geq 1, \end{cases}$$

This comes out of the “periodic” resolution of $\mathbf{Z}$ as a trivial G -module:

$$\dots \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\varepsilon} \mathbf{Z} \rightarrow 0,$$

where $N = 1 + \sigma + \dots + \sigma^{n-1}$ is the norm map and ε sends σ to 1. A similar computation holds for the infinite cyclic group $G = \mathbf{Z}$:

$$H^0(\mathbf{Z}, A) = A^{\mathbf{Z}}, \quad H^1(\mathbf{Z}, A) = A/(1 - \sigma)A, \quad H^i(\mathbf{Z}, A) = 0 \text{ for all } i \geq 2.$$

For a subgroup $H \subset G$, we denote by

$$M_H^G(A) := \text{Hom}_H(\mathbf{Z}[G], A)$$

the G -module with action

$$\sigma(\phi)(g) = \phi(\sigma^{-1}g), \quad \sigma \in G, g \in G, \phi \in M_H^G(A).$$

Then, *Shapiro's lemma* tells us that there is a canonical isomorphism

$$H^i(G, M_H^G(A)) \cong H^i(H, A). \quad (3.1)$$

3.1 Restriction, Inflation and Corestriction

Let H be a subgroup of G . The *restriction* map is defined on cocycles as

$$\text{Res} : H^i(G, A) \rightarrow H^i(H, A), \quad \text{Res}([f]) = [f|_H],$$

which explains the name: $f|_H$ is just restriction to H .

For a normal subgroup $H \triangleleft G$, the quotient G/H acts on A^H and we have the *inflation* map

$$\text{Inf} : H^i(G/H, A^H) \rightarrow H^i(G, A), \quad \text{Inf}([f]) = [f \circ \pi],$$

where $\pi : G \rightarrow G/H$ is the canonical projection.

The *corestriction* map goes in the opposite direction of restriction, and is defined as follows: let $H \subset G$ be a subgroup of finite index. This is well-defined and gives a map

$$\text{Cor} : H^i(H, A) \rightarrow H^i(G, A).$$

3.2 Profinite groups

Let $G = \varprojlim G_\alpha$ be a profinite group (that is, an inverse limit of finite groups). This is a topological group, whose topology is induced by the discrete topology on the finite groups G_α .

We have natural continuous maps $\pi_\alpha : G \rightarrow G_\alpha$, and their kernels U_α form a system of open neighborhoods of $1 \in G$.

Remark 3.2.

- We may assume that the G_α coincide with **all** finite quotients of G by open normal subgroups, as if Γ is such a quotient, then there is some α such that U_α has trivial image in Γ : indeed, Γ is finite and the intersection of all U_α is trivial. So, we get a factorization

$$\begin{array}{ccc} G & & \\ \downarrow & \searrow \pi_\alpha & \\ G_\alpha & \dashrightarrow & \Gamma \end{array}$$

- The inverse limit of a system of finite groups is compact and totally disconnected. With compact here we also include the Hausdorff property, which holds by Tychonoff's theorem. Moreover, the open subgroups of G are exactly the closed subgroups of finite index.

3.3 Cohomology of profinite groups

Let $G = \varprojlim G_\alpha$ be a profinite group, and let A be a *continuous* G -module, i.e. an abelian group with an action of G by automorphisms such that $\forall a \in A$, the stabilizer G_a is open in G (exercise: this is equivalent to the map $G \times A \rightarrow A$ being continuous, where A has the discrete topology). In particular, if $G_\alpha = G/U_\alpha$, then A^{U_α} is a G_α -module, and we define:

Definition 3.3.

$$H_{\text{cont}}^i(G, A) := \varinjlim H^i(G_\alpha, A^{U_\alpha}),$$

where the limit is taken with respect to the inflation maps induced by the projections $G_\beta \twoheadrightarrow G_\alpha$:

$$H^i(G_\alpha, A^{U_\alpha}) \rightarrow H^i(G_\beta, A^{U_\beta}).$$

We'll immediately drop the subscript *cont*, writing $H^i(G, A)$.

In the case where $G = \text{Gal}(k^s|k)$ is the absolute Galois group of a field k , we'll use the notation $H^i(k, A)$ for $H^i(G, A)$.

Proposition 3.4. If G is profinite and A is a continuous G -module, then $H^i(G, A)$ is a torsion abelian group for all $i > 0$.

Proof. The groups $H^i(G_\alpha, A^{U_\alpha})$ are torsion: they are killed by the order of G_α . Therefore, their direct limit $H^i(G, A)$ is also torsion. $\square$

Corollary 3.5. If moreover A is a $\mathbf{Q}$ -vector space, then $H^i(G, A) = 0$ for all $i > 0$.

Remark 3.6. Given a finite group G and a finitely generated G -module A , the cohomology groups $H^i(G, A)$ are finite for all $i > 0$. This follows from the description of cohomology in terms of the standard resolution and the torsion property above.

Lemma 3.7. Let G be profinite with (U_α) a system of open normal subgroups. Then,

- If H is a closed subgroup of G , there is a canonical isomorphism

$$H^i(H, A) \cong \varinjlim H^i(H/(H \cap U_\alpha), A^{H \cap U_\alpha})$$

which shows that H is profinite;

- If moreover H is normal in G , then $G/H \cong \varprojlim G/(HU_\alpha)$ is profinite.

Therefore, we can define restriction, corestriction and inflation maps for profinite groups by taking the direct limit of the corresponding maps for finite groups: for instance, restriction is induced by the maps

$$H^i(G/U_\alpha, A^{U_\alpha}) \xrightarrow{\text{res}} H^i(H/(H \cap U_\alpha), A^{H \cap U_\alpha}).$$

Since we defined cohomology explicitly as a direct limit, we have to show “by hand” the existence of the long exact sequence in cohomology associated with a short exact sequence of G -modules $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$.

Lemma 3.8. Let $(A_\alpha, \phi_{\alpha\beta})$, $(B_\alpha, \psi_{\alpha\beta})$ and $(C_\alpha, \rho_{\alpha\beta})$ be direct systems of G_α -modules, and assume given short exact sequences

$$0 \rightarrow A_\alpha \xrightarrow{f_\alpha} B_\alpha \xrightarrow{g_\alpha} C_\alpha \rightarrow 0$$

for all α which are compatible with the transition maps (i.e. morphisms of direct systems). Then, the direct limit of these sequences is exact.

The non-trivial part is defining the connecting homomorphism

$$\delta : H^i(G, C) \rightarrow H^{i+1}(G, A).$$

Let $U \subset G$ be an open normal subgroup. If $K_U := \text{coker}(B^U \rightarrow C^U)$, then $\varinjlim_U K_U = 0$. By the lemma, we have a short exact sequence

$$\varinjlim H^i(G/U, (B^U/A^U)) \rightarrow \varinjlim H^i(G/U, C^U) \rightarrow \varinjlim H^{i+1}(G/U, K_U)$$

and the last term is 0 by the above. Therefore, for small enough U we can lift a cocycle $f \in Z^i(G, C)$ to some element in $\varinjlim H^i(G/U, (B^U/A^U))$, and then apply the connecting homomorphism coming from the short exact sequence $0 \rightarrow A^U \rightarrow B^U \rightarrow B^U/A^U \rightarrow 0$ to get an element in $\varinjlim H^{i+1}(G/U, A^U)$, which has a map into $H^{i+1}(G, A)$ (of course, one should check that this does not depend on any of the choices made).

4 Kummer Theory

Proposition 4.1 (Hilbert's Theorem 90). Let $K|k$ be a finite Galois extension with group $G = \text{Gal}(K|k)$. Then

$$H^1(G, K^\times) = 0.$$

Consequently, by passing to the limit over all such extensions we get

$$H^1(k, (k^s)^\times) = 0.$$

Lemma 4.2 (Dedekind). If $\sigma_1, \dots, \sigma_n$ are the elements of G , then they are linearly independent over K .

Proof. Suppose they are linearly dependent, and let

$$a_1\sigma_1 + \dots + a_n\sigma_n = 0$$

be the *shortest* nontrivial linear relation. Then, choosing $x \in K$ such that $\sigma_1(x) \neq \sigma_2(x)$ and $y \in K$ is arbitrary, then

$$\begin{aligned} 0 &= a_1\sigma_1(xy) + \dots + a_n\sigma_n(xy) \\ &= a_1\sigma_1(x)\sigma_1(y) + \dots + a_n\sigma_n(x)\sigma_n(y) \\ &= \sigma_1(x)(a_1\sigma_1(y) + \dots + a_n\sigma_n(y)) + (\sigma_2(x) - \sigma_1(x))(a_2\sigma_2(y) + \dots + a_n\sigma_n(y)). \end{aligned}$$

Putting this together with the original relation evaluated at y and multiplied by $\sigma_1(x)$, we get a shorter relation

$$(\sigma_2(x) - \sigma_1(x))(a_2\sigma_2(y) + \dots + a_n\sigma_n(y)) = 0,$$

which is a contradiction. □

Proof of Proposition. Let $\varphi : G \rightarrow K^\times$ be a 1-cocycle, with the cocycle condition being $\varphi(\sigma\tau) = \varphi(\sigma) \cdot \sigma(\varphi(\tau))$. We want to show that φ is a coboundary, i.e. there exists some $y \in K^\times$ such that $\varphi(\tau) = \tau(y) \cdot y^{-1}$ for all $\tau \in G$. Consider the element

$$\Phi = \sum_{\sigma \in G} \varphi(\sigma) \cdot \sigma \in K[G].$$

By the lemma, Φ is non-zero. Therefore there exist $x, y \in K^\times$ such that

$$y = \Phi(x) = \sum_{\sigma \in G} \varphi(\sigma) \cdot \sigma(x),$$

and applying τ to both sides gives

$$\tau(y) = \sum_{\sigma \in G} \tau\varphi(\sigma) \cdot \tau\sigma(x) = \sum_{\sigma \in G} \frac{\varphi(\tau\sigma)}{\varphi(\tau)} \tau\sigma(x) = \frac{1}{\varphi(\tau)} \sum_{\sigma \in G} \varphi(\tau\sigma) \cdot \tau\sigma(x) = \frac{1}{\varphi(\tau)} \sum_{\sigma \in G} \Phi(x) = \frac{y}{\varphi(\tau)},$$

□

The main result of this section is the following:

Proposition 4.3. Let k be a field and m a positive integer which is prime to $\text{char}(k)$ (this implies that $\mu_m \subset k^s$). Then, there is a canonical isomorphism

$$H^1(k, \mu_m) \cong k^\times / k^{\times m}.$$

whose inverse sends the class of $a \in k^\times / k^{\times m}$ to the class of the cocycle $\sigma \mapsto \sigma(a) \cdot a^{-1}$, where $\alpha \in (k^s)^\times$ is any m -th root of a .

Proof. Consider the short exact sequence of G_k -modules

$$1 \rightarrow \mu_m \rightarrow (k^s)^\times \xrightarrow{(\cdot)^m} (k^s)^\times \rightarrow 1.$$

By Hilbert's Theorem 90, we have $H^1(k, (k^s)^\times) = 0$. Therefore, the long exact sequence in cohomology gives

$$H^0(k, (k^s)^\times) \xrightarrow{(\cdot)^m} H^0(k, (k^s)^\times) \rightarrow H^1(k, \mu_m) \rightarrow 0,$$

and the cokernel of the first map is exactly $k^\times / k^{\times m}$, since $H^0(k, (k^s)^\times) = (k^s)^{\times G_k} = k^\times$. The last statement then follows by writing out the connecting map explicitly (lifting corresponds to taking an m -th root...). □

Example 4.4. If moreover $\mu_m \subset k$, then $H^1(k, \mu_m) = H^1(k, \mathbf{Z}/m\mathbf{Z}) = \text{Hom}(G_k, \mathbf{Z}/m\mathbf{Z})$ (the action is trivial). In this case, we deduce that every finite extension $K|k$ with $\text{Gal}(K|k) \cong \mathbf{Z}/m\mathbf{Z}$ is of the form $K = k(\alpha)$ with $\alpha^m \in k$.

5 Cup-products

Let G be a group and A, B be G -modules. The module $A \otimes_{\mathbf{Z}} B$ is a G -module with the diagonal action $\sigma(a \otimes b) = \sigma(a) \otimes \sigma(b)$. The cup-product is a bilinear pairing

$$\begin{aligned} \cup : H^i(G, A) \times H^j(G, B) &\rightarrow H^{i+j}(G, A \otimes B) \\ (a, b) &\mapsto a \cup b \end{aligned}$$

which is functorial in A, B and graded-commutative: $a \cup b = (-1)^{ij} b \cup a$; there will be a version for profinite groups by passing to the limit. Let us sketch the application of this gadget to the Hilbert symbol:

Definition 5.1. For a field k , the *Brauer group* of k is

$$\mathrm{Br}(k) := H^2(k, k_s^\times).$$

A non-trivial result is that elements of $\mathrm{Br}(k)$ correspond to central division algebras which are finite-dimensional over k .

Another piece of the Kummer long exact sequence is

$$H^1(k, k_s^\times) = 0 \rightarrow H^2(k, \mu_n) \xrightarrow{(\cdot)^n} H^2(k, k_s^\times),$$

from which we get $H^2(k, \mu_n) \cong {}_n\mathrm{Br}(k)$, the n -torsion in the Brauer group.

Suppose that $\mu_n \subset k$, and take $a, b \in k^\times$. Then, we can take their classes in $k^\times/k^{\times n} \cong H^1(k, \mu_n) = H^1(k, \mathbf{Z}/n\mathbf{Z})$, and take their cup-product:

$$H^1(k, \mathbf{Z}/n\mathbf{Z}) \times H^1(k, \mathbf{Z}/n\mathbf{Z}) \rightarrow H^2(k, \mathbf{Z}/n\mathbf{Z} \otimes_{\mathbf{Z}} \mathbf{Z}/n\mathbf{Z}) \cong {}_n\mathrm{Br}(k).$$

Theorem 5.2 (Hasse). Let $k|\mathbf{Q}_p$ be a finite extension. Then,

$$\mathrm{Br}(k) \cong \mathbf{Q}/\mathbf{Z}.$$

In particular, we deduce that for any n ${}_n\mathrm{Br}(k) \cong \frac{1}{n}\mathbf{Z}/\mathbf{Z}$. Putting these together, we get a map

$$(\cdot, \cdot)_n : k^\times/k^{\times n} \times k^\times/k^{\times n} \rightarrow H^1(k, \mathbf{Z}/n\mathbf{Z}) \times H^1(k, \mathbf{Z}/n\mathbf{Z}) \rightarrow {}_n\mathrm{Br}(k) \cong \mathbf{Z}/n\mathbf{Z},$$

which is the *Hilbert symbol*. What happens if $\mu_n \not\subset k$? Given a character $\chi : G_k \rightarrow \mathbf{Z}/n\mathbf{Z}$, this lives in the first cohomology group $H^1(k, \mathbf{Z}/n\mathbf{Z})$, so we can take its cup-product with the class of any $b \in k^\times$ inside $H^1(k, \mu_n)$:

$$(\chi, b) := \chi \cup b \in H^2(k, \mu_n) \xrightarrow{\text{Hasse}} \mathbf{Z}/n\mathbf{Z}.$$

Construction 5.3. Let $A^\bullet, B^\bullet$ be complexes of e.g. abelian groups. The tensor product of

these complexes is defined as follows: consider the double complex

$$\begin{array}{ccccccc}
& \vdots & & \vdots & & \vdots & \\
& \uparrow & & \uparrow & & \uparrow & \\
A^{i-1} \otimes B^{j+1} & \longrightarrow & A^i \otimes B^{j+1} & \longrightarrow & A^{i+1} \otimes B^{j+1} & \longrightarrow & \dots \\
& \uparrow & & \uparrow & & \uparrow & \\
A^{i-1} \otimes B^j & \longrightarrow & A^i \otimes B^j & \longrightarrow & A^{i+1} \otimes B^j & \longrightarrow & \dots \\
& \uparrow & & \uparrow & & \uparrow & \\
A^{i-1} \otimes B^{j-1} & \longrightarrow & A^i \otimes B^{j-1} & \longrightarrow & A^{i+1} \otimes B^{j-1} & \longrightarrow & \dots
\end{array}$$

with horizontal and vertical differentials

$$\partial_{i,j}^h = \partial_A^i \otimes \text{id}_{B^j}, \quad \partial_{i,j}^v = (-1)^i \text{id}_{A^i} \otimes \partial_B^j.$$

With this sign, the diagram anti-commutes, i.e. $\partial_{i,j+1}^h \circ \partial_{i,j}^v = -\partial_{i+1,j}^v \circ \partial_{i,j}^h$. Recall that the total complex $T(A^\bullet \otimes B^\bullet)$ in this case is defined as

$$T^n(A^\bullet \otimes B^\bullet) = \bigoplus_{i+j=n} A^i \otimes B^j,$$

with differential given on each component by the sum of the horizontal and vertical differentials:

$$\partial^n = \sum_{i+j=n} \partial_{i,j}^h + \partial_{i,j}^v.$$

Because of the anti-commutativity, we have $\partial^{n+1} \circ \partial^n = 0$, which shows that this is a complex.

Definition 5.4. We denote by $(A \otimes B)^\bullet$ the total complex $T(A^\bullet \otimes B^\bullet)$, and call it the *tensor product* of the two complexes.

Assume moreover that A, B are abelian groups, and consider the complexes $\text{Hom}(A^\bullet, A)$ and $\text{Hom}(B^\bullet, B)$, with differentials given by pre-composition with ∂_A, ∂_B . We want to define a product

$$H^i(\text{Hom}(A^\bullet, A)) \times H^j(\text{Hom}(B^\bullet, B)) \rightarrow H^{i+j}(\text{Hom}((A \otimes B)^\bullet, A \otimes B)).$$

If $\alpha \in \text{Hom}(A^i, A)$ and $\beta \in \text{Hom}(B^j, B)$, then $\alpha \otimes \beta : A^{-i} \otimes B^{-j} \rightarrow A \otimes B$ and we get a map

$$\text{Hom}(A^{-i} \otimes B^{-j}, A \otimes B) \rightarrow \text{Hom}\left(\bigoplus_{k+l=i+j} A^{-k} \otimes B^{-l}, A \otimes B\right) = \text{Hom}(T^{i+j}(A^\bullet \otimes B^\bullet), A \otimes B),$$

and one easily checks that this is compatible with differentials (sends cocycles to cocycles and coboundaries to coboundaries). Therefore, it induces such a product on cohomology, which is the desired cup-product.

Lemma 5.5. Let $P_\bullet \rightarrow \mathbf{Z}$ be a projective resolution of $\mathbf{Z}$ as a trivial G -module, and recall the homological indexing $P_i = P^{-i}$. Then, $(P \otimes P)_\bullet$ is a projective resolution of $\mathbf{Z}$ as a trivial $(G \times G)$ -module, where the action is the diagonal one.

Applying the previous construction with $A^\bullet = B^\bullet = P_\bullet$, we get a product

$$H^i(\mathrm{Hom}_G(P_\bullet, A)) \times H^j(\mathrm{Hom}_G(P_\bullet, B)) \rightarrow H^{i+j}(\mathrm{Hom}_{G \times G}((P \otimes P)_\bullet, A \otimes B)),$$

and composing with the map $H^{i+j}(\mathrm{Hom}_{G \times G}((P \otimes P)_\bullet, A \otimes B)) \rightarrow H^{i+j}(\mathrm{Hom}_G((P \otimes P)_\bullet, A \otimes B))$ induced by the diagonal embedding $G \hookrightarrow G \times G$, we get a cup-product

$$H^i(G, A) \times H^j(G, B) \rightarrow H^{i+j}(G, A \otimes B).$$

This is compatible with restriction, inflation and corestriction maps: if $H \leq G$ is a subgroup, then

- For $a \in H^i(G, A)$ and $b \in H^j(G, B)$, we have $\mathrm{Res}(a \cup b) = \mathrm{Res}(a) \cup \mathrm{Res}(b) \in H^{i+j}(H, A \otimes B)$.
- If $H \triangleleft G$ is normal, then for $a \in H^i(G/H, A^H)$ and $b \in H^j(G/H, B^H)$, we have $\mathrm{Inf}(a \cup b) = \mathrm{Inf}(a) \cup \mathrm{Inf}(b) \in H^{i+j}(G, A \otimes B)$.
- If $[G : H] < \infty$, then for $a \in H^i(H, A)$ and $b \in H^j(G, B)$, we have

$$\mathrm{Cor}(a \cup \mathrm{Res}(b)) = \mathrm{Cor}(a) \cup b \in H^{i+j}(G, A \otimes B)$$

which is called the *projection formula*.

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